

Flow Separation Criterion

Continuity Equation (curvilinear)

$$\frac{1}{1+Kn} \frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} + \frac{K}{1+Kn} (\ell u_n) = 0$$

At wall, no slip and no permeability.

$$(u_s, u_n) \Big|_{n=0, \text{wall}} = (0, 0)$$

→ At $n=0$, wall, s = any value in the domain

$$u_s(s, n=0) \Rightarrow u_s(s) \Big|_{n=0} = 0$$

$$u_n(s, n=0) \Rightarrow u_n(s) \Big|_{n=0} = 0$$

Due to no slip

$$\text{Therefore, } \frac{\partial u_s}{\partial s}(s) \Big|_{n=0} = 0 \text{ and } \frac{\partial u_n}{\partial s}(s) \Big|_{n=0} = 0$$

$$\rightarrow \left(\frac{1}{1+K(0)} \frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} + \frac{K}{1+K(0)} (\ell u_n) = 0 \right)_{n=0}$$

$$\left(\frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} = 0 \right)_{n=0}$$

$$\left(\ell \frac{\partial u_s}{\partial s} + u_s \frac{\partial \ell}{\partial s} + \ell \frac{\partial u_n}{\partial n} + u_n \frac{\partial \ell}{\partial n} = 0 \right)_{n=0}$$

$$\therefore \left(\ell \frac{\partial u_n}{\partial n} \right)_{n=0} = 0; \ell > 0 \text{ at all } s$$

$$\text{Therefore, } \frac{\partial u_n}{\partial n} \Big|_{n=0} = 0 \text{ at wall for all } s$$

Flow Separation Criterion

Streamwise/tangential momentum equation (curvilinear)

$$\rho \left(\frac{u_s}{1+Kn} \frac{\partial u_s}{\partial s} + \frac{\partial u_s}{\partial n} + \frac{K u_s u_n}{1+Kn} \right) = - \frac{1}{1+Kn} \frac{\partial P}{\partial s} + \frac{1}{1+Kn} \frac{\partial \tau_{ss}}{\partial s}$$

$$0 = - \frac{\partial P}{\partial s} + \frac{\partial \tau_{ss}}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K \tau_{sn} \text{ at wall} + \frac{\partial \tau_{sn}}{\partial n} + \frac{2K}{1+Kn} \tau_{sn}$$

$$\tau_{ss} = \frac{4}{3} \mu \left(\frac{1}{1+Kn} \frac{\partial u_s}{\partial s} + \frac{K u_n}{1+Kn} \right) - \frac{2}{3} \mu \frac{\partial u_n}{\partial n} \text{ at wall } (n=0, u_s=0, u_n=0)$$

$$\frac{\partial \tau_{ss}}{\partial s} = \frac{4}{3} \mu \left(\frac{1}{1+Kn} \frac{\partial^2 u_s}{\partial s^2} + \frac{\partial u_s}{\partial s} \frac{\partial}{\partial s} \left(\frac{1}{1+Kn} \right) + \frac{\partial u_n}{\partial s} \left(\frac{K}{1+Kn} \right) + \frac{\partial}{\partial s} \left(\frac{K}{1+Kn} \right) \right)$$

$$u_s(s) \Big|_{n=0} = 0 \rightarrow \frac{\partial u_s(s)}{\partial s} \Big|_{n=0} = 0 \rightarrow \frac{\partial^2 u_s(s)}{\partial s^2} \Big|_{n=0} = 0 - \frac{2}{3} \mu \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0}$$

$$u_n(s) \Big|_{n=0} = 0 \rightarrow \frac{\partial u_n(s)}{\partial s} \Big|_{n=0} = 0 \left[\frac{\partial \tau_{ss}}{\partial s} \right] = - \frac{2}{3} \mu \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0}$$

$$\frac{\partial u_n(s)}{\partial n} \Big|_{n=0} = 0 \rightarrow \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0} = 0$$

$$\therefore \frac{\partial \tau_{ss}}{\partial s} = - \frac{2}{3} \mu(0) \rightarrow \frac{\partial \tau_{ss}}{\partial s} = 0$$

$$\therefore \left(0 = - \frac{\partial P}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K \tau_{sn} \right) \Big|_{n=0, \text{ wall}}$$

Flow Separation Criterion

$$\tau_{sn} = \mu \left(\frac{\partial u_s}{\partial n} - \frac{K u_s}{1+Kn} + \frac{1}{1+Kn} \frac{\partial u_n}{\partial s} \right)$$

$u_s=0$ $\frac{\partial u_n}{\partial s}=0$

$$\tau_{sn} = \mu \frac{\partial u_s}{\partial n} \Big|_{n=0} \text{ at wall because}$$

$$\therefore \left(0 = -\frac{\partial p}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K\mu \frac{\partial u_s}{\partial n} \right) \Big|_{n=0} \quad \begin{matrix} u_s(s)=0 \text{ and } u_n(s)=0 \\ \text{and } \frac{\partial u_n(s)}{\partial s} = 0 \end{matrix}$$

$$\frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - \frac{K}{1+Kn} \frac{\partial u_s}{\partial n} - \frac{1}{1+Kn} \frac{\partial}{\partial n} \left(\frac{\partial u_n}{\partial s} \right) + \frac{1}{1+Kn} \frac{\partial^2 u_n}{\partial n \partial s} \right) \Big|_{n=0}$$

$$\therefore \frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - \frac{K}{1+K(0)} \frac{\partial u_s}{\partial n} + \frac{1}{1+K(0)} \frac{\partial^2 u_n}{\partial n \partial s} \right) \Big|_{n=0} + \frac{\partial u_n}{\partial s} \Big|_{n=0} \frac{\partial}{\partial n} \left(\frac{1}{1+Kn} \right) \Big|_{n=0}$$

$$\frac{\partial^2 u_n}{\partial n \partial s} = \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right); \quad \frac{\partial^2 u_n}{\partial n \partial s} (s, n=0) = \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \Big|_{n=0} \right) = 0$$

$$\frac{\partial u_n}{\partial n} (s) \Big|_{n=0} = 0 \rightarrow \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} (s) \Big|_{n=0} \right) = 0$$

$$\rightarrow \left(\frac{\partial \tau_{sn}}{\partial n} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - K \frac{\partial u_s}{\partial n} \right) \right) \Big|_{n=0}$$

$$\frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0} - K\mu \frac{\partial u_s}{\partial n} \Big|_{n=0}$$

Flow Separation Criterion

$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \left.\frac{\partial^2 u_s}{\partial n}\right|_{n=0} + 2K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} - K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0} + 2K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} + K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\boxed{\left.\frac{\partial p}{\partial s}\right|_{n=0} = \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} + K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}}$$

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Flow Separation Criterion

Since $\left[\frac{\partial p}{\partial s} \right]_{n=0} = \mu \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0} + K \mu \frac{\partial u_s}{\partial n} \Big|_{n=0}$

Thin Boundary Layer Assumption

$\delta \ll L$. Assume thin boundary layer before flow separates

$$u_s(n=\delta) = u_s(n=0) + \delta \frac{\partial u_s}{\partial n} \Big|_{n=0} + \frac{\delta^2}{2} \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0} + \dots$$

$u_{s,e}$ Taylor Series Approximation

$$\therefore u_{s,e} = \delta \frac{\partial u_s}{\partial n} \Big|_{n=0} + \frac{\delta^2}{2} \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0} \quad (2)$$

$$\rightarrow \left[\frac{\partial p}{\partial s} \right]_{n=0} - K \mu \frac{\partial u_s}{\partial n} \Big|_{n=0} = \mu \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0}$$

$$\frac{1}{\mu} \left[\frac{\partial p}{\partial s} \right]_{n=0} - K \frac{\partial u_s}{\partial n} \Big|_{n=0} = \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0}$$

$$\frac{1}{\mu} \left[\frac{\partial p}{\partial s} \right]_{n=0} - \frac{1}{R} \frac{\partial u_s}{\partial n} \Big|_{n=0} = \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0}$$

Flow Separation Criterion

Since

$$\frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0} = \frac{1}{\mu} \frac{\partial p}{\partial s} \Big|_{n=0} - \frac{1}{R} \frac{\partial u_s}{\partial n} \Big|_{n=0}$$

and since $u_{s,e} = \delta \frac{\partial u_s}{\partial n} \Big|_{n=0} + \frac{\delta^2}{2} \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0}$

Substitute

$$u_{s,e} = \delta \frac{\partial u_s}{\partial n} \Big|_{n=0} + \frac{\delta^2}{2} \left(\frac{1}{\mu} \frac{\partial p}{\partial s} \Big|_{n=0} - \frac{1}{R} \frac{\partial u_s}{\partial n} \Big|_{n=0} \right)$$

$$u_{s,e} = \left(\delta - \frac{\delta^2}{2R} \right) \frac{\partial u_s}{\partial n} \Big|_{n=0} + \frac{\delta^2}{2\mu} \frac{\partial p}{\partial s} \Big|_{n=0}$$

$$2\mu u_{s,e} = \left(2\delta - \frac{\delta^2}{R} \right) \mu \frac{\partial u_s}{\partial n} \Big|_{n=0} + \delta^2 \frac{\partial p}{\partial s} \Big|_{n=0}$$

$$\frac{2\mu u_{s,e}}{2\delta - \frac{\delta^2}{R}} - \frac{\delta^2}{2\delta - \frac{\delta^2}{R}} \frac{\partial p}{\partial s} \Big|_{n=0} = \mu \frac{\partial u_s}{\partial n} \Big|_{n=0}$$

$$\frac{2\mu R u_{s,e}}{2R\delta - \delta^2} - \frac{R\delta^2}{2R\delta - \delta^2} \frac{\partial p}{\partial s} \Big|_{n=0} = \mu \frac{\partial u_s}{\partial n} \Big|_{n=0}$$

Flow Separation Criterion

$$\therefore \left[\frac{2\mu R u_{s,e}}{2R\delta - \delta^2} - \frac{R\delta}{2R - \delta} \frac{\partial p}{\partial s} \right]_{n=0} = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

Before flow separation

within thin boundary layer approximation

Before flow separation

$$\rightarrow \left[\frac{2\mu R u_{s,e}}{2R\delta - \delta^2} - \frac{R\delta}{2R - \delta} \frac{\partial p}{\partial s} \right]_{n=0} = \tau_{wall}$$

Flow separates when the flow exerts zero shear stress on the wall and shear stress sign reverse and becomes negative ($\tau_{wall} \leq 0$)

$$\left[\frac{2\mu R u_{s,e}}{2R\delta - \delta^2} - \frac{R\delta}{2R - \delta} \frac{\partial p}{\partial s} \right]_{n=0} = \tau_{wall} \leq 0$$

Flow Separation Criterion

$$\therefore \frac{2\mu R u_{s,e}}{2R\delta - \delta^2} - \frac{R\delta}{2R - \delta} \frac{\partial P}{\partial s} \bigg|_{n=0} \leq 0$$

$$\frac{R\delta}{2R - \delta} \frac{\partial P}{\partial s} \bigg|_{n=0} \geq \frac{2\mu R u_{s,e}}{2R\delta - \delta^2}$$

is the criteria in order for
flow to separate

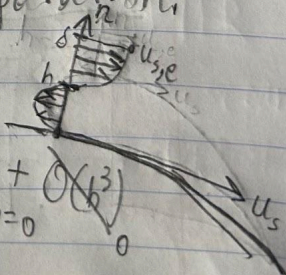
Thin Recirculation Bubble Assumption

h is partial height in normal units (n) from the wall where tangential/streamwise velocity u_s becomes 0 from negative in the recirculation bubble that occurs AFTER flow separation.

Assume thin, small $h \ll L$

Taylor Series Approximation

$$u_s(n=h) = u_s(n=0) + h \frac{\partial u_s}{\partial n} \bigg|_{n=0} + \frac{h^2}{2} \frac{\partial^2 u_s}{\partial n^2} \bigg|_{n=0} + O(h^3)$$



Flow Separation Criterion

$$\therefore 0 = h \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \frac{h^2}{2} \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0}$$

$$0 = \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \frac{h}{2} \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0}$$

$$0 = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \frac{h\mu}{2} \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0}$$

$$\text{Since } \left. \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} + K \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} \quad (1)$$

$$\downarrow$$

$$\left. \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} + \frac{\mu}{R} \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$\mu \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} = \left. \frac{\partial p}{\partial s} \right|_{n=0} - \frac{1}{R} \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$\rightarrow 0 = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \frac{h}{2} \left(\left. \frac{\partial p}{\partial s} \right|_{n=0} - \frac{1}{R} \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} \right)$$

Flow Separation Criterion

$$\therefore 0 = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \frac{h}{2} \left(\left. \frac{\partial p}{\partial s} \right|_{n=0} - \frac{1}{R} \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} \right)$$

$$0 = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \frac{h}{2} \left. \frac{\partial p}{\partial s} \right|_{n=0} - \frac{h}{2R} \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$0 = \left(1 - \frac{h}{2R} \right) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \frac{h}{2} \left. \frac{\partial p}{\partial s} \right|_{n=0}$$

$$- \frac{h}{2} \left. \frac{\partial p}{\partial s} \right|_{n=0} = \left(1 - \frac{h}{2R} \right) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$\frac{-h/2}{1 - h/(2R)} \left. \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$\left. - \frac{hR}{2R-h} \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

or

$$\left. - \frac{hR}{2R-h} \frac{\partial p}{\partial s} \right|_{n=0} = \tau_{wall}$$

AFTER
Flow separation,
in separation
bubble

Flow Separation Criterion

At/After flow separation $\tau_{\text{wall}} < 0$

Since $-\frac{hR}{2R-h} \frac{\partial p}{\partial s} \bigg|_{n=0} = \tau_{\text{wall}} < 0$

$$\boxed{-\frac{hR}{2R-h} \frac{\partial p}{\partial s} \bigg|_{n=0} < 0}$$

or

$$\boxed{\frac{2hR}{2R-h} \frac{\partial p}{\partial s} \bigg|_{n=0} > 0}$$

At separated
flow in
recirculation
bubble.